

AGE 316: MICROCOMPUTER SYSTEMS

LECTURE 13: COMMUNICATION PROTOCOLS (I^2C & SPI)

Dr. Kadeghe Fue

Department of Agricultural Engineering

Recap: How devices talk

Lecture 7: Serial (UART)

- Uses 2 wires: **TX** and **RX**.
- **Asynchronous:** No clock wire. Speed (Baud Rate) must be agreed upon.
- **Limitation:** Can only connect 2 devices (1-to-1).

Problem: What if we want to connect 10 sensors to one Arduino? We run out of pins!

Concept: Synchronous Comm

To connect many devices efficiently, we use a **Clock Line**.

The Clock

A wire that pulses ON/OFF regularly (like a metronome).

Sync

Data is sent only when the clock ticks. No need to guess the speed.

Master/Slave

One boss (Arduino) controls the clock; everyone else listens.

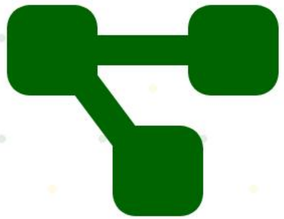
Protocol 1: I²C

Inter-Integrated Circuit

Also known as "Two-Wire Interface" (TWI).

- **SDA (Serial Data):** Data flows back and forth here.
- **SCL (Serial Clock):** Keeps time.

Superpower: You can connect up to **127 devices** using just these TWO wires!



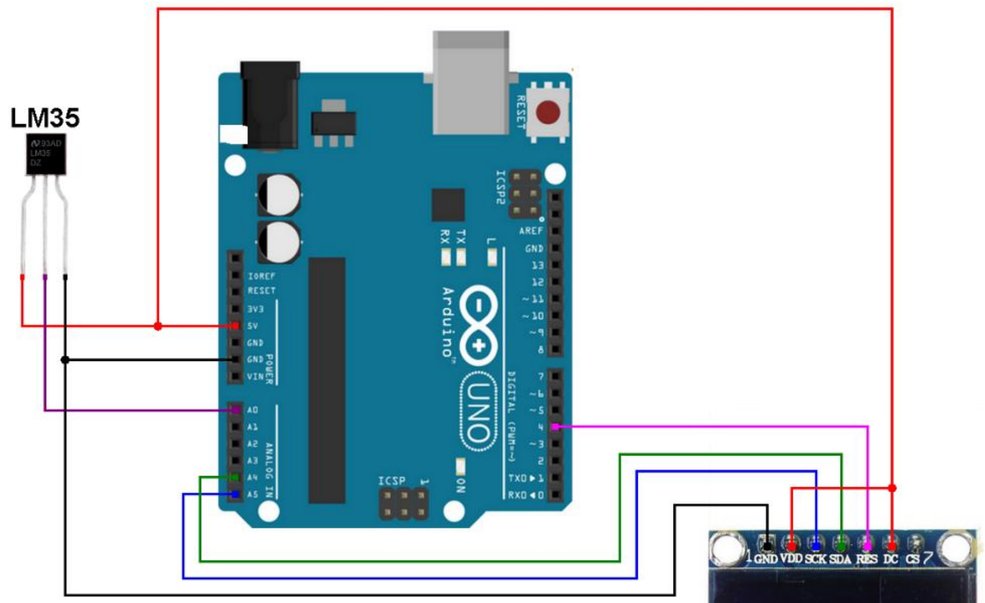
How I²C Works: Addressing

If everyone shares the same wire, how do we know who is talking?



Every device has a unique **HEX Address** (e.g., 0x27 for LCD, 0x68 for RTC). The Master shouts "Hey 0x68!" and only that device replies.

Wiring I²C



SSD1306 OLED
128x64 Pixel

On Arduino Uno:

- **SDA:** Analog Pin A4
- **SCL:** Analog Pin A5

⚠ **Note:** I2C lines are "Open Drain". They often need 4.7kΩ Pull-up resistors to 5V (though many modules include them).

Code: The Wire Library

Arduino uses the `Wire.h` library for I2C.

```
#include

void setup() {
  Wire.begin(); // Join bus as Master

  Wire.beginTransmission(0×68); // Talk to RTC
  Wire.write(0); // Send data
  Wire.endTransmission(); // Stop
}
```

Protocol 2: SPI

Serial Peripheral Interface

Designed for **SPEED**.

Used for high-data devices like SD Card readers, TFT screens, and Cameras.

Trade-off: Uses 4 wires instead of 2.



Very Fast

Wiring SPI

MOSI

Master Out
Slave In

Pin 11

MISO

Master In
Slave Out

Pin 12

SCK

Serial
Clock

Pin 13

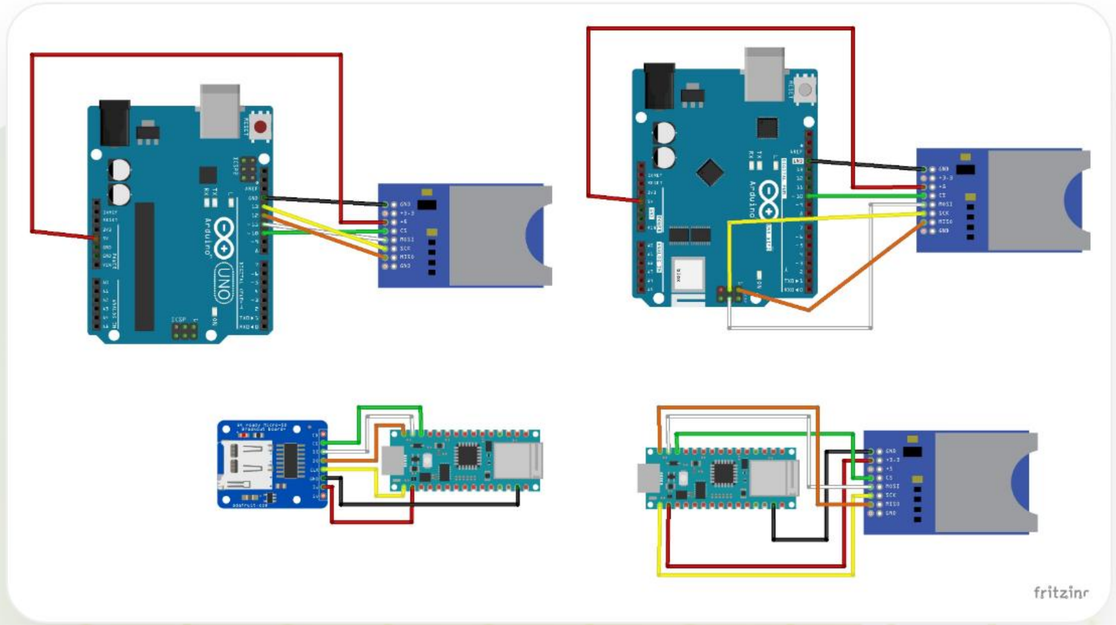
CS / SS

Chip
Select

Any Pin (10)

How SPI Works: Chip Select

SPI doesn't use addresses. It uses a dedicated wire for every slave.



- To talk to Device A, Master pulls Device A's **CS Pin LOW**.
- Device B ignores everything because its CS pin is HIGH.
- **Benefit:** Extremely fast switching.
- **Drawback:** More devices = More wires.

Code: The SPI Library

```
#include

void setup() {
  pinMode(10, OUTPUT); // CS Pin
  SPI.begin();
}

void loop() {
  digitalWrite(10, LOW); // Wake up Slave
  SPI.transfer(100); // Send data
  digitalWrite(10, HIGH); // Sleep Slave
}
```

Comparison

Feature	UART (Serial)	I2C	SPI
Wires	2 (TX, RX)	2 (SDA, SCL)	4 (MOSI, MISO, SCK, CS)
Speed	Slow	Medium	Very Fast
Devices	1-to-1	Up to 127	Limited by Pins
Pros	Simple, Debugging	Saves Pins	High Speed

Applications in Agriculture

Sensors (I2C)

Temp/Humidity (SHT31), Light (BH1750), Pressure (BMP280). Easy to daisy-chain.

Data Logging (SPI)

SD Card Modules use SPI to write data fast enough so we don't lose readings.

Displays

OLEDs usually use I2C for simplicity.

Safety: Logic Levels

Warning

Arduino Uno operates at **5V**.

Many modern I2C/SPI sensors (like SD Cards) operate at **3.3V**.

Connecting 5V to a 3.3V device can destroy it instantly.

The Solution

Use a **Logic Level Converter** (Bi-Directional).

Or check if your module has an onboard regulator (often looks like a small black chip).

Summary

Synchronous

Uses a clock line to keep timing perfect.

I2C

2 Wires. Uses Addresses. Good for sensors.

SPI

4 Wires. Uses Chip Select. Good for SD Cards/Displays.

Lab Prep: Sensor Network

Next Lab:

- ✓ We will connect an **I2C LCD**.
- ✓ We will connect an **I2C Temperature Sensor** (e.g., BMP280).
- ✓ We will display the temperature on the screen using the I2C bus.

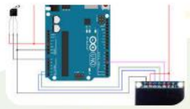
Questions?

"The only stupid question is the one you don't ask."

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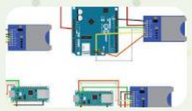
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Image Sources



https://simple-circuit.com/wp-content/uploads/2019/03/arduino-lm35-ssd1306-i2c-oled-display.png

Source: simple-circuit.com



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